

Please write at least 50 characters about what you learned in today's lecture.

Today I learned that time series data is unique because chronological order matters, a property known as temporal ordering. We covered feature engineering techniques like moving averages to smooth data, and cyclical encoding with trigonometric functions to capture periodic structures. For modeling, we explored STL decomposition to split data into trend, seasonal, and residual components. I also learned how to use the ADF test to check for stationarity, and how to forecast using statistical models like AR, MA, ARMA, and ARIMA. Lastly, we learned to use the Akaike Information Criterion (AIC) to balance model complexity and accuracy when picking hyperparameters.

Please tell us specifically what you found particularly helpful in today's lecture.

I really found the Exploratory Data Analysis (EDA) section in the Colab notebooks helpful. Applying STL decomposition and plotting correlograms for the sunspot dataset, which has an 11-year cycle, made the concepts of seasonality and autocorrelation very easy to visualize. Seeing the step-by-step transformation of the non-stationary Hawaii atmospheric CO2 concentration data into stationary data using differentiation was also a great practical example.

Please let us know if there were any parts that were unclear or if there are any areas for improvement.

The mathematical formulas for the statistical models were a bit overwhelming at first. Also, during the first notebook session, there was a minor error: the method returned a resampler object instead of a dataframe, which caused a brief moment of confusion. However, I really appreciated that Shinsuke took the time to consult the Pandas documentation and explain the fix to us clearly.

Please tell us specifically about the instructor, including what you liked and what you were dissatisfied with.

I really liked how reassuring Shinsuke was during the Q&A session. When a student commented that the topic was too hard and they felt lost, he kindly acknowledged that time series is naturally one of the most difficult concepts for beginners and encouraged us to just take our time reviewing it. That was very comforting.

Please write down any lectures or subjects you would like to see offered in the future.

Since we focused on classical statistical models today, I would love to learn more about machine learning and deep learning approaches for time series that were briefly mentioned, such as RNNs, LSTMs, and Transformers. Also, I am very excited about next week's guest lectures by Mr. Shakamoria on student entrepreneurship and Mr. Yuki on AI implementation across industries.

Please feel free to write your comments.

I'm so relieved that there is no homework today, as it gives me the time I need to properly review these complex models. Also, a huge congratulations to the students at Brack University in Bangladesh for being chosen to host the final in-person session and the one-day hackathon.